GHXTox: Peptide toxicity prediction by fusing a task-adaptive protein language model with pLDDT-aware multi-scale conformer ensembles

LaLaLa, LaLaLa, LaLaLa, LaLaLa

Abstract

As biomedical research advances from the cellular to the molecular level, peptides have become increasingly important as a versatile class of bioactive molecules and are now widely used in the targeted treatment of diseases such as Parkinson's disease and cancer. However, toxicity assessment remains a critical bottleneck in peptide-based drug development. Existing methods have begun to explore multi-dimensional (multimodal) feature fusion and toxicity prediction based on the three-dimensional (3D) conformations of peptides, yet most of them still underutilize 3D structural information, and systematic multi-dimensional fusion remains rare. Here, we propose GHXTox, a multi-dimensional fusion framework that combines a one-dimensional (1D) sequence branch with a 3D structure branch. The 1D branch analyzes sequences with a task-adaptive protein language model (ProtT5 fine-tuned with low-rank adaptation, LoRA) and a bidirectional Mamba state-space model. The 3D branch extracts multi-scale structural information from predicted conformations, from local residue functional groups to the whole peptide, and exploits the predicted local distance difference test (pLDDT) confidence through quality-aware gating. GHXTox achieves competitive performance on independent test sets and provides more detailed biological interpretations for peptide toxicity prediction.

Keywords: peptide toxicity prediction; multimodal deep learning; protein language model; bidirectional Mamba; pLDDT-aware conformer ensemble; interpretability

1 INTRODUCTION

As biomedical research advances from the cellular to the molecular level, peptides have become increasingly important as a versatile class of bioactive molecules and are now widely used in the targeted treatment of diseases such as Parkinson's disease and cancer [1]. Peptide toxicity is a critical safety consideration throughout the drug development pipeline, and the accurate and efficient identification of toxic peptides remains a major bottleneck in peptide-based drug discovery [2]. Because conventional experimental toxicity assays are costly, time-consuming, and difficult to scale, reliable computational predictors are highly desirable.

Computational peptide toxicity prediction has evolved from traditional machine learning methods based on handcrafted physicochemical features toward deep learning approaches with stronger representation-learning capability. Current mainstream methods can be roughly grouped into two categories. The first category operates on sequence-level (one-dimensional, 1D) and/or atomic-level (two-dimensional, 2D) representations. For example, ToxMSRC combines continuous bag-of-words (CBOW) embeddings, multi-scale convolutional neural networks, and a bidirectional long short-term memory network with residual connections, achieving strong performance on the benchmark dataset [2]. ToxPLTC integrates pooled ProtT5 embeddings, borderline-SMOTE augmentation, and a text convolutional neural network [3]. ToxiPep fuses context-aware sequence representations with atomic-level graphs through a cross-attention mechanism [4]. HyPepTox-Fuse further combines embeddings from multiple protein language models

with conventional descriptors [5]. Although these models achieve strong performance within their respective modalities, they largely neglect the three-dimensional (3D) conformations of peptides, which are closely related to toxicity; cross-dimensional information interaction and mutual validation remain limited, and their interpretability is mostly confined to the sequence level.

The second category attempts to exploit 3D structural information. PeptiTox uses ESM2 sequence embeddings together with ESMFold-predicted 3D structures and geometric graph neural networks [6]. StrucToxNet feeds ProtT5 sequence embeddings and ESMFold-predicted structures into an equivariant graph neural network (EGNN) and systematically analyzes structural quality [7]. These studies demonstrate the value of 3D conformations for toxicity prediction. However, such methods are constrained by the limited accuracy of current structure prediction models on short peptides and typically rely on a single predicted conformation; structural confidence, i.e., the predicted local distance difference test (pLDDT) score, is either discarded through hard filtering or ignored, and multi-conformer ensembles are not considered. Moreover, the fusion between the sequence and structure branches remains relatively shallow, and no systematic multi-dimensional fusion that combines a task-adaptive sequence model with confidence-aware multi-scale structure modeling has been established.

To address these limitations, we propose GHXTox, an interpretable multi-dimensional fusion framework for peptide toxicity prediction that combines a 1D sequence branch with a 3D structure branch. In the 1D branch, a task-adaptive protein language model (ProtT5 fine-tuned with low-rank adaptation, LoRA) is combined with a bidirectional Mamba state-space model for sequence analysis [8–10]. In the 3D branch, because current structure prediction models generally output a single predicted conformation with limited accuracy on short peptides, we adopt a pragmatic structure augmentation scheme in which different structure prediction and conformational sampling models are applied across different sequence-length ranges [11–14]. On this basis, we design a multi-scale structural information extraction strategy that progressively captures structural cues from local residue functional groups (chemical sites) to the whole peptide, and the pLDDT confidence output by structure prediction models is exploited through continuous down-weighting and dynamic gating of structural features. The two branches are then fused with weights determined on training out-of-fold predictions, followed by probability calibration. Ablation experiments demonstrate that the two dimensions complement each other effectively and that the fusion substantially improves overall performance, while interpretability analyses provide more detailed biological explanations for the predictions.

In summary, the main contributions of this work are threefold: (1) we propose GHXTox, an interpretable multi-dimensional fusion framework that integrates a task-adaptive 1D protein language model with pLDDT-aware multi-scale 3D structure modeling; (2) the 3D branch designs a multi-scale structural information extraction strategy spanning from local residue functional groups to the whole peptide, and incorporates pLDDT confidence into structural feature processing through continuous down-weighting and dynamic gating; and (3) extensive experiments on benchmark and independent test sets show that GHXTox achieves competitive performance, and ablation, robustness, and interpretability analyses confirm the effectiveness and complementarity of the two dimensions.

2 MATERIALS AND METHODS

2.1 Datasets

We evaluate GHXTox on the benchmark datasets constructed by CAPTP [17] and subsequently adopted by ToxMSRC [2]. Following the data construction protocol of ToxMSRC [2], toxic peptide sequences were merged from three databases, namely CSM-Toxin [18], ToxinPred2 [19], and ATSE [20], and sequences longer than 50 residues were removed. Non-toxic sequences were retrieved from UniProt [21] using the query “NOT KW-0800 AND NOT KW-0020 AND

reviewed:true", and sequences containing non-standard amino acids or exceeding 50 residues were excluded. After combining the two groups and removing samples with conflicting labels, CD-HIT [22] was applied with a sequence-identity threshold of 0.9 to eliminate redundant sequences, yielding 2,138 toxic and 5,375 non-toxic peptides. Following the same partition as ToxMSRC [2], we used 85% of the data as the training set and the remaining 15% as the independent Test1 set, resulting in 1,818 toxic and 4,569 non-toxic training peptides and 320 toxic and 806 non-toxic Test1 peptides. An additional independent Test2 set, containing 46 toxic and 536 non-toxic peptides, was used to further evaluate generalization ability. Table **??** summarizes the statistics of the datasets used in this study.

Table 1: Statistics of the datasets used in this study.

Dataset	Toxic	Non-toxic	Total	Use
Training (benchmark)	1,818	4,569	6,387	Model training
Test1 (benchmark)	320	806	1,126	Independent test
Test2	46	536	582	Independent test

2.2 Overview of the Model

Figure 1: Placeholder: overall framework of the proposed model (mirrors Figure 1 of the reference paper; replace with your own flowchart and caption).

2.2.1 ProtT5 Semantic Embedding

2.2.2 Secondary Structure Embedding

2.2.3 Gated Cross-Attention Mechanism

2.2.4 Multi-Scale Gated Convolution Network

2.2.5 Output Layer

2.3 Fast Gradient Method (FGM) for Robustness Evaluation

2.4 Evaluation Criteria

3 RESULTS AND DISCUSSION

3.1 Performance Comparison with Existing Models

Table 2: Placeholder: performance comparison with existing models (mirrors Table 2 of the reference paper; replace with your own).

Figure 2: Placeholder: performance comparison of different pre-trained protein language models (mirrors Figure 2 of the reference; replace with your own).

Table 3: Placeholder: comparison of projection dimensions, kernel sizes and attention heads (mirrors Table 3 of the reference paper; replace with your own).

3.2 Exploration of the Model Architecture

3.3 Ablation Experiments

3.3.1 Analysis of the Effectiveness of Multimodal Features

3.3.2 Contribution Analysis of the Cross-Modal Fusion Mechanism

3.4 Evaluation of Model Robustness and Generalization

3.4.1 Robustness Analysis Under Embedding and Sequence-Level Perturbations

3.4.2 Analysis of the Model's Generalization Ability on the Imbalanced Test Sets

3.5 Visualization of Feature Learning Using UMAP

3.6 Interpretability Analysis

3.6.1 Feature Importance Analysis

3.6.2 Attention-Based Interpretability Analysis

3.6.3 Mutation-Scan Analysis

3.7 Limitations and Future Directions

4 CONCLUSIONS

Data and Software Availability

Author Contributions

Acknowledgements

Conflict of Interest Disclosure

References

Figure 3: Placeholder: results of the ablation study (mirrors Figure 3 of the reference paper; replace with your own).

Table 4: Placeholder: complexity and performance comparison (mirrors Table 4 of the reference paper; replace with your own).

Table 5: Placeholder: additional complexity introduced by the added modules (mirrors Table 5 of the reference paper; replace with your own).

Table 6: Placeholder: K-fold cross-validation results (mirrors Table 6 of the reference paper; replace with your own).

Table 7: Placeholder: robustness under FGM adversarial perturbation (mirrors Table 7 of the reference paper; replace with your own).

Table 8: Placeholder: robustness under conservative substitution perturbations (mirrors Table 8 of the reference paper; replace with your own).

Table 9: Placeholder: prediction-probability stability under perturbations (mirrors Table 9 of the reference paper; replace with your own).

Table 10: Placeholder: comparison on the imbalanced independent test set (mirrors Table 10 of the reference paper; replace with your own).

Table 11: Placeholder: comparison on the external imbalanced test set (mirrors Table 11 of the reference paper; replace with your own).

Figure 4: Placeholder: feature visualization of different model stages (mirrors Figure 4 of the reference paper; replace with your own).

Figure 5: Placeholder: SHAP feature-importance analysis (mirrors Figure 5 of the reference paper; replace with your own).

Figure 6: Placeholder: SHAP-based contribution analysis (mirrors Figure 6 of the reference paper; replace with your own).

Figure 7: Placeholder: attention-based interpretability analysis (mirrors Figure 7 of the reference paper; replace with your own).

Figure 8: Placeholder: mutation-scan analysis (mirrors Figure 8 of the reference paper; replace with your own).